

A PROJECT REPORT ON

**Unlocking Clinical Data: Structuring And
Summarizing Unstructured Medical Records**

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ABSTRACT

Using the BERT algorithm, we describe a robust system that summarizes medical data and disambiguates abbreviations in the ever-expanding world of clinical documentation, where every second counts and clarity might be a lifeline. Without sacrificing important detail, our implementation uses BERT's deep contextual knowledge to extract important information from verbose medical records and condense it into brief, insightful summaries. Concurrently, the model accurately decodes context-sensitive terms like "RA" (Rheumatoid Arthritis vs. Right Atrium) and resolves ambiguities in domain-specific acronyms. The purpose of this dual-task pipeline is to assist medical professionals, lessen cognitive overload, and facilitate quicker, better-informed decision-making. Compared to baseline models, our results show better medical accuracy, coherence, and relevancy. Our results demonstrate improved coherence, relevancy, and medical accuracy when compared to baseline models. In essence, this work is a step toward more intelligent, human-centric healthcare NLP systems, blending efficiency with empathy in the language of medicine.

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CHAPTER 1

INTRODUCTION

1.1 Overview

The widespread adoption of Electronic Health Record (EHR) systems has transformed modern healthcare by digitizing patient records and enabling seamless sharing of clinical information. These systems enhance clinical decision-making by allowing healthcare providers to easily access medical histories, diagnoses, prescriptions, and treatment data. However, despite these benefits, EHRs face significant challenges—especially when handling unstructured clinical data such as physician notes, discharge summaries, or test results. A major source of complexity in processing such data is the widespread use of abbreviations, which are often ambiguous and context-dependent.

For instance, the abbreviation “HR” may refer to “heart rate,” “hormone receptor,” or “healthcare resources” depending on the specialty and context. Similarly, a condition like heart disease can be referred to using various terms, such as “myocardial infarction,” “ischemia,” or “coronary artery disease.” These linguistic inconsistencies, along with disorganized document structure, frequent typographical errors, and the critical need to maintain patient privacy, make automated analysis of clinical texts a difficult task.

Traditional approaches such as rule-based systems, pattern-matching, and early machine learning techniques have attempted to address these issues with limited success. More recently, deep learning methods have been applied to Natural Language Processing (NLP) tasks in the biomedical domain, achieving better performance in areas such as named entity recognition (NER), part-of-speech tagging, and semantic analysis. However, these systems often struggle to generalize due to the domain-specific nature of medical language and the wide variety of data formats found in EHRs.

This project proposes an implementation-focused solution that aims to standardize unstructured clinical inputs—specifically, handwritten or free-text patient diagnoses, prescriptions, and clinical forms—by transforming them into structured digital records. The primary focus is on the disambiguation of clinical abbreviations using a combination of NLP techniques and machine learning models. By resolving abbreviation ambiguity, the system enhances clarity and interpretability of clinical data, making it more usable for both healthcare providers and biomedical researchers.

The proposed methodology begins with data collection and preprocessing, where a large and diverse corpus of clinical texts is cleaned and tokenized to prepare it for analysis. In the abbreviation detection phase, a hybrid approach is used—combining dictionary lookup methods with morphology-based pattern recognition—to identify abbreviations in the text. Contextual feature extraction follows, leveraging syntactic and semantic relationships to understand the surrounding context of each abbreviation.

To determine the correct meaning of each abbreviation, a machine learning-based disambiguation model is trained on labeled clinical data. Features

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such as surrounding words, sentence structure, and word embeddings (e.g., Word2Vec or BERT) are used to train classifiers like SVMs or deep learning models. Human validation is incorporated in the post-processing stage to ensure accuracy, particularly for high-risk terms. Finally, the system's performance is measured using standard evaluation metrics like precision, recall, F1-score, and validated through cross-validation techniques to ensure robustness.

Through this approach, the project contributes a scalable framework for transforming ambiguous and unstructured clinical data into clear, standardized formats. This work supports more effective use of EHR data in patient care, clinical workflows, and medical research.

1.2 Motivation

Clinical abbreviations have become a double-edged sword in the fast-paced world of modern medicine, where notes are written quickly, and seconds can save lives. They speed up documentation while also obscuring clarity. Electronic Health Records (EHRs) are a tangle of mysterious codes because a single acronym can have multiple meanings depending on the situation, area, or specialty.

In addition to being confusing, this lack of standardization causes misunderstandings, delays in diagnosis, and even medical mistakes. Decoding this shorthand chaos is crucial as AI and NLP become more prevalent. In the healthcare industry, resolving abbreviations is more than just a technical task; it's about bridging the gap between human expression and machine understanding. Making every note matter, every document readable, and every choice a bit more accurate is the goal.

1.3 Problem Definition and Objectives

Clinical notes in Electronic Health Records (EHRs) often contain abbreviations that vary by context, specialty, and region, making them difficult to interpret accurately. This ambiguity creates challenges for NLP systems in extracting meaningful information, leading to reduced accuracy in clinical decision support and data analysis. Resolving these abbreviations is essential to ensure clarity, consistency, and reliability in healthcare data processing. Specifically, the project seeks to:

- Identify and extract abbreviations from unstructured clinical text within Electronic Health Records (EHRs).
- Develop context-aware models that can accurately expand abbreviations based on medical domain and usage.
- Construct or integrate a comprehensive medical abbreviation dictionary aligned with various clinical specialties.
- Compare the performance of traditional rule-based, statistical, and deep learning methods for abbreviation resolution.
- Integrate the abbreviation resolution system as a module within a broader NLP pipeline for improved EHR data interpretation.

1.4 Project Scope & Limitations

Project Scope:

This project focuses on resolving clinical abbreviations found in unstructured text within Electronic Health Records (EHRs). It involves the extraction of abbreviations, context-aware disambiguation, and accurate expansion using a combination of rule-based, statistical, and deep learning techniques. The scope includes building or integrating a medical abbreviation dictionary, developing and evaluating models on clinical datasets, and implementing the solution as part of a larger NLP pipeline. The project is limited to text-based data and does not include real-time clinical deployment or integration with live hospital systems.

Limitations:

Despite the widespread implementation of Electronic Health Record (EHR) systems, which have greatly improved access to and management of patient data, a substantial portion of critical clinical information remains locked within unstructured textual formats. These include handwritten notes, progress reports, discharge summaries, and diagnosis descriptions. Unlike structured fields such as lab results or vital signs, unstructured data is not readily usable by automated systems and requires complex processing to extract meaningful insights.

One of the primary challenges in utilizing this unstructured data is the widespread use of abbreviations by healthcare professionals. These abbreviations are often context-dependent and can carry multiple interpretations. For instance, the abbreviation "HR" could signify "heart rate" in a cardiology context, "hormone receptor" in oncology, or "healthcare resources" in administrative contexts. Without proper contextual analysis, such ambiguities can result in misinterpretation, leading to errors in diagnosis, treatment decisions, or data analysis for research purposes.

Additional challenges associated with the use of unstructured clinical text in EHRs include:

Inconsistency in Documentation Styles: Different physicians or departments may document medical information in varied formats, making it difficult to apply uniform extraction or interpretation methods.

- **Complex Medical Language:** Clinical texts often include highly specialized terminology, shorthand notations, domain-specific acronyms, and incomplete sentences. These complexities pose significant difficulties for natural language processing (NLP) systems.
- **Noise and Irrelevant Data:** Handwritten notes and digitized forms may include spelling mistakes, special characters, or non-informative content

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(e.g., HTML tags or template text), which can interfere with automated analysis.

- **Contextual Disambiguation:** Accurately interpreting an abbreviation or term depends heavily on the surrounding context, such as the specialty of the clinician, the section of the document, or other nearby medical terms. Without proper context modeling, automated systems can select incorrect interpretations.
- **Data Privacy and Anonymization:** Any system working with patient records must account for patient privacy and comply with data protection regulations. De-identifying sensitive information without losing relevant clinical content is itself a complex task.
- **Lack of Labeled Data:** Training machine learning models for abbreviation expansion and disambiguation requires annotated datasets that pair abbreviations with their correct meanings in context. Such datasets are scarce, particularly across different medical domains and languages.
- **Scalability Across Specialties:** Abbreviations and documentation styles vary significantly between medical specialties (e.g., cardiology, oncology, neurology), requiring adaptable models that can generalize across domains.
- **Integration with Existing EHR Systems:** Any proposed system must be capable of integrating with various EHR platforms, which differ in data schema, user interfaces, and operational workflows.

By addressing the challenges like “Inconsistency in Documentation Styles, Noise and Irrelevant Data”, the proposed project aims to build a system that not only transforms unstructured clinical texts into structured, analyzable data but also resolves abbreviation ambiguities through NLP and machine learning techniques. This advancement will significantly enhance the usability of EHR data, reduce potential errors in interpretation, and unlock valuable information that can contribute to both clinical decision-making and medical research.

1.5 Methodologies of Problem solving

To address the challenges of ambiguity and verbosity in clinical records, the project adopted a two-stage NLP approach combining both rule-based and deep learning methods.

1. Abbreviation Disambiguation:

Medical abbreviations were first identified using morphological patterns and a curated dictionary. For context-sensitive interpretation, a BERT-based model was used to predict the correct expansion based on surrounding words, effectively resolving ambiguity in medical terms.

2. Text Summarization:

The cleaned and disambiguated reports were passed to a transformer-based summarization model, *google/pegasus-xsum*. This model was selected for its strong performance in generating concise, abstractive summaries of long-form text.

3. Preprocessing and Data Handling:

All inputs were normalized and truncated to meet model constraints. PyTorch's DataLoader and tokenization pipelines ensured consistent input formatting.

4. Training and Optimization:

Mixed-precision training (*torch.cuda.amp*) and gradient accumulation were used to overcome memory limitations. Model checkpoints were regularly saved to Google Drive to support resumption and evaluation.

5. Integration and Testing:

Components were integrated into a unified pipeline allowing user input, abbreviation resolution, summarization, and output display. Each module was tested independently and then as part of the end-to-end flow.

This structured methodology ensured the system was accurate, modular, and adaptable to future extensions like domain-specific fine-tuning or multilingual support.

CHAPTER 2

LITERATURE SURVEY

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The paper [1] by Essam et al. explains the use of Machine Learning in Medical Science. It addresses the challenges like ambiguity in terminology, the importance of domain-specific machine learning models, and the scarcity of labeled datasets. Moreover, it discusses the advancement in technology to improve performance in biomedical NLP. Various ML algorithms are used, and their application in tasks like document classification and named entity recognition.

The paper [2] by Temitope Ibrahim Amosa et al. focuses on the issue of errors arising due to the use of ambiguous abbreviations in EHR. Key methodologies stated are:

- Machine Learning Approaches which include Utilization of supervised and unsupervised algorithms like Support Vector Machines and Neural Networks.
- Contextual Analysis that examines surrounding text to infer the meaning of acronyms.
- Word Sense Disambiguation (WSD) techniques to determine the correct meaning of acronyms based on context.
- Feature Extraction to identify relevant features from text data for improved disambiguation.
- Natural Language Understanding (NLU) that leverages broader context to enhance accuracy.
- Evaluation Metrics includes use of precision, recall, and F1-score to assess accuracy.

The paper [3] by Hui Yang et al. presents a system developed for the i2b2 Obesity Challenge, aimed at predicting the status of obesity and 15 related comorbidities from patient discharge summaries. The system tackled two tasks: (1) a textual task, identifying explicit disease mentions, and (2) an intuitive task, inferring disease status from indirect evidence. Using a hybrid approach combining dictionary look-up, rule-based techniques, and machine learning, the system achieved a macro-averaged F-measure of 81% for the textual task (the highest in the challenge) and 63% for the intuitive task (ranking 7th). With accuracy comparable to human annotators, the study demonstrates the potential of text mining to improve medical data analysis and disease status prediction.

The paper [4] by Seyedmostafa Sheikhalishahi et al. gives review of Natural Language Processing techniques applied to clinical notes for chronic disease management. Chronic diseases like diabetes and heart disease generate vast amounts of unstructured clinical data in electronic health records (EHRs), which makes NLP a key tool for extracting valuable information. This paper provides a summary of the current state of NLP techniques used to process clinical notes related to chronic diseases along with key challenges and advancements. The methods they reviewed are namely entity recognition (NER),

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relation extraction, and machine learning methods. Despite significant advancements, challenges remain in processing noisy, unstructured data and ensuring interoperability across systems.

In the paper [5] by Anoual El Kah and Imad Zeroual explores the application of Natural Language Processing in analyzing Electronic Health Records. It highlights the substantial increase in EHR usage, leading to a vast amount of unstructured language data related to patient care. As EHR data is unstructured hence it complicates use of traditional data analysis due to complexities of human language so NLP techniques like Named Entity Recognition and relation extraction are emphasized that convert unstructured text into machine-readable formats.

In their paper [6], Long Chen et al. present an advanced NLP system designed to extract key details on medications and adverse drug events (ADEs) from clinical narratives. It focuses on the applications of combining deep learning techniques with domain based medical knowledge to increase the accuracy of the model. The authors adopt a hybrid approach, integrating traditional knowledge-based methods with deep learning to enhance the system's ability to identify medications and ADEs. By leveraging medical ontologies, the system captures nuanced relationships that might be missed by standalone methods. Tested on a large dataset of clinical notes, the system showed notable improvements in extraction accuracy compared to traditional approaches, demonstrating the effectiveness of combining knowledge-based and machine learning techniques for processing healthcare data.

In paper [7], Amila Kugic et al. conducted a systematic review in order to analyze studies on processing short-form clinical content with the help of NLP and Machine learning. Thematic analysis was used by them so as to categorize findings based on techniques, effectiveness, and applications. They pick out methods to mark incomplete and ambiguous data in clinical text.

Samantha Williams and Elena Petrovich [8] discuss how techniques like text summarization, entity extraction, and machine learning comprehension in NLP can draw out insights from unstructured healthcare data, giving accuracy over 90%. They not only focused attention on advancements like transfer learning and graph-based representation but also focused on important challenges like ethical concerns, interoperability and security.

In paper [9], Emre Sezgin et al. focuses on recognition of medical entities like symptoms and treatments by analyzing patient-generated text, using NLP. This study showed the potential of NLP in extracting medical insights; though future enhancement is needed for advanced implementation. Also, this method demonstrated challenges and promises included maintaining accuracy and managing language variability over health domains.

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Method	Entity Recognition	Abbreviation Expansion	Scalability	Performance	Resource Requirements
CRF	Moderate	Poor	High	Moderate	Low
BioBERT	High	High	Moderate	Excellent	High
Rule-Based	Low	High	High	Varies	Low
SciBERT	High	High	Moderate	Excellent	High
CNN	Moderate	Poor	Moderate	Good	Medium
Transformer XL	High	High	Moderate	Excellent	High
BERT+CRF	High	Moderate	Moderate	Excellent	High
Hybrid (ML+Rule-Based)	Moderate	High	Moderate	Good	Low to Medium

Tab 2.1: Comparative Analysis

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

Assumptions:

- The input to the model is structured or semi-structured medical text (e.g., symptoms, vitals, diagnosis).
- The summarization model (like BioBART, BioGPT, or BART) is hosted and accessible through an API endpoint (e.g., <http://localhost:5000/summarize>).
- Frontend and backend are either hosted on the same origin or CORS is enabled on the backend to handle cross-origin requests.
- The model is not concurrently optimized for heavy load; it is assumed to serve one request at a time in a demo or small-scale setting.
- Input text is in English and reasonably well-formatted.
- If using Hugging Face models, the machine requires internet access, or the model must be pre-downloaded.
- It is assumed that sufficient system memory and compute (CPU/GPU) are available to load and run the model.

Dependencies:

- The performance of the system relies on the quality and comprehensiveness of the datasets used for training. The classification model cannot handle or generalize to document types not included in the training data.
- The system depends on the use of pre-trained deep learning models and requires adequate tuning and adaptation to the specific long document classification tasks at hand.
- High-performance computing resources, such as cloud GPUs, are required for both training and deploying the model efficiently, as the project involves deep learning-based techniques which can be computationally intensive.
- The classification outcomes are dependent on the data preparation and preprocessing steps. If the input data is not well-structured, it can adversely affect the model's classification accuracy.
- The ability of the system to handle large-scale documents is contingent on its training with extensive, diverse datasets that represent the variety of long documents it will be required to classify.

3.2 Functional Requirements

The functional requirements define the key operations of the system.

3.2.1 Medical Text Input and Summarization

Description: The system shall allow users to input detailed medical data related to a patient and receive a concise, medically accurate summary using a transformer-based model.

- The system shall provide a user interface to input medical details such as symptoms, vitals, diagnosis, and medications
- The frontend shall validate the input for non-empty values before submission.
- The system shall send a POST request to the backend with the input data in JSON format.
- The backend shall process the input using a medical summarization model (e.g., BioBART, BioGPT).
- The backend shall return a summarized response in less than 5 seconds for a single input (under normal load).
- The frontend shall display the summarized text clearly for the user.

3.2.2 Model Access and API Integration

Description: The system shall expose a REST API for summarizing medical records using pretrained models and ensure smooth communication between frontend and backend.

- The backend shall expose an endpoint at /summarize that accepts POST requests with input text.
- The system shall initialize the transformer model at runtime and keep it in memory for quick access.
- The backend shall support CORS to allow frontend integration from different origins.
- The system shall log each request and its corresponding summary for monitoring and debugging.

- The backend shall handle invalid inputs and return appropriate error messages.
- The system shall ensure that the summarization model returns outputs within defined length limits (e.g., 30–50 words).

3.2.3 Abbreviation Detection and Disambiguity Resolvance

Description: The system shall identify medical abbreviations in the input data and resolve their full meanings to ensure clarity and accuracy in the summarization process.

- The system shall scan the input medical text for known abbreviations (e.g., "BP", "HR", "SOB").
- The backend shall maintain a dictionary of standard medical abbreviations and their full forms.
- Upon detecting an abbreviation, the system shall replace it with its full form before sending it to the summarization model.
- The system shall support contextual disambiguation of abbreviations based on surrounding words (e.g., "MS" could be "Mitral Stenosis" or "Multiple Sclerosis").
- If an abbreviation has multiple possible meanings, the system shall apply simple NLP techniques or rules to infer the correct expansion.
- The final output shall be free from unresolved or ambiguous abbreviations to improve readability for healthcare providers and patients.

3.3 Nonfunctional Requirements

3.3.1 Performance Requirements

- For inputs under 512 tokens, the system must produce summarized findings from medical text inputs in 3 seconds.
- At least ten concurrent queries should be handled by the summarization model without noticeably degrading performance.
- 99% system uptime must be maintained during a 30-day rolling window.
- Under typical load, the average response time of the frontend-to-backend API cannot be more than 500 milliseconds.

3.3.2 Safety Requirements

- To let consumers know that the summaries are not a substitute for expert medical advice, the system must have disclaimers.
- The model is not allowed to recommend diagnoses or make clinical judgements on its own.
- The system will report a temporary unavailability notification in the event of a model failure or unresponsiveness.
- Logs and error messages must not contain any sensitive information that can be used to identify a patient.

3.3.3 Security Requirements

- HTTPS encryption must be used for all data transfers between the frontend and backend.
- All input and output must be processed in-memory unless specifically requested and anonymized; patient data cannot be kept in a permanent location.
- The use of token-based authentication will safeguard the API.
- Depending on the deployment region, the system must abide by applicable data protection regulations like GDPR or HIPAA.
- To keep a secure runtime environment, regular vulnerability scans and dependency upgrades must be planned.

3.3.4 Software Quality Attributes

- **Reliability:** Under various input loads, the system must reliably produce correct summaries.
- **Maintainability:** The codebase will adhere to modular design principles, making model replacements and updates simple.
- **Scalability:** Using orchestration tools like Kubernetes, the backend infrastructure must be containerized and prepared for horizontal scaling.
- **Usability:** The frontend should provide researchers and clinicians with a simple, uncluttered interface that includes output formatting and clear instructions.
- **Portability:** With few setup adjustments, the system should operate on local servers and cloud services like AWS and GCP.

3.4 System Requirements

3.4.1 Database Requirements

- The system shall use a structured database to store electronic health records (EHR), patient summaries, and abbreviation mappings.
- A relational database such as PostgreSQL or MySQL is recommended for robust querying and integrity constraints.
- The database shall support storage of text fields (e.g., patient descriptions, summaries) and numerical values (e.g., vitals, lab results).
- The system shall ensure data privacy through appropriate encryption and access control measures.
- The database schema shall include tables for :
 - Patient Information
 - Medical History
 - Abbreviations Dictionary
 - Summarized Reports

3.4.2 Software Requirements (Platform Choice)

- Backend:
 - Programming Language: Python 3.8+
 - Frameworks: Flask or FastAPI(for RESTful API integration)
 - ML Libraries: HuggingFace Transformers, PyTorch or TensorFlor
 - NLP Models: BioBERT, BioBART or similar domain-specific models
- Frontend:
 - Technologies: React.js(preferred), HTML5, CSS3
 - State Management: Redux or Context API(if needed)
- Environment:
 - Jupyter or Google Colab(for model development)

- Docker (for containerization and deployment)

3.4.3 Hardware Requirements

- The system requires minimum: 8 GB RAM, 2.4 GHz processor, 50 GB storage.

3.5 Analysis Models: SDLC Model to be applied

The Agile Software Development Life Cycle (SDLC) approach will be used for the creation of this Electronic Health Record Summarization System because of its flexibility, iterative nature, and emphasis on user feedback and ongoing improvement. The Agile model's justification and implementation in this situation are provided below:

- Incremental Delivery: Working components such as data creation, a summarization API, and a frontend interface will be delivered in phases as the system is developed in brief, manageable sprints.
- Regular Evaluation: To improve the precision of the summarization model and the usability of the user interface, ongoing input from domain experts (such as test users or medical professionals) will be obtained.
- Rapid Prototyping: To allow for real-time testing, early deployments of the ML model (for example, using BioBERT) and backend APIs will be made available.
- Adaptive Planning: Any sprint can include requirements changes, for as adding more disease categories or supporting more NLP activities (like disambiguation).
- Cross-functional Collaboration: To make sure the system is operational, user-friendly, and clinically beneficial, developers, data scientists, and UI/UX designers will coordinate their efforts.
- Chance Reduction: By gradually resolving difficulties, early iterations lower the chance of model inaccuracies, performance problems, or usability issues.

Because the needs for this project are always changing, particularly with regard to machine learning performance tuning, user interface design, and frontend and backend system integration, agile is a good fit.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The system is designed to address two significant issues in Electronic Health Records (EHR): the presence of ambiguous medical abbreviations and the complexity of unstructured clinical notes. The design is modular, enabling quick processing, contextual comprehension, and real-time summarization.

4.1.1 Input Module

First, the system receives digitized clinical documents, including handwritten prescriptions, diagnosis notes, and discharge summaries.

4.1.2 Preprocessing Pipeline

Text Cleaning: Removal of noise including HTML tags, special characters, and irrelevant formatting.

Tokenization: Breaking down sentences into tokens (words, abbreviations, phrases).

Anonymization: De-identification of personally identifiable information (PII) to comply with privacy standards.

4.1.3 Abbreviation Detection & Extraction

Dictionary-based Lookup: This method flags frequently used acronyms by using a carefully managed medical abbreviation dictionary.

Morphology-based Rules: Uses part-of-speech and regex patterns to identify abbreviation patterns.

4.1.4 Contextual Feature Extraction

Utilizes dependency parsing and syntactic analysis to extract each abbreviation's local and global context.

Makes use of domain-aware tagging, which determines if a sentence belongs in the section on prescriptions or diagnoses.

4.1.5 Disambiguation Module

Uses BERT or refined transformer models (ClinicalBERT or SciBERT, optional) to figure out how to expand an acronym according to the surrounding content.

4.1.6 Summarization Module

Verbose text is condensed into comprehensible summaries via a sequence-to-sequence transformer (like Pegasus). Clinicians benefit from this output because it improves decision support and lessens their reading load.

4.1.7 Validation and Output

Human-in-the-loop checks for cases that pose a significant risk.
Produces structured data in formats that are compatible with most EHRs.

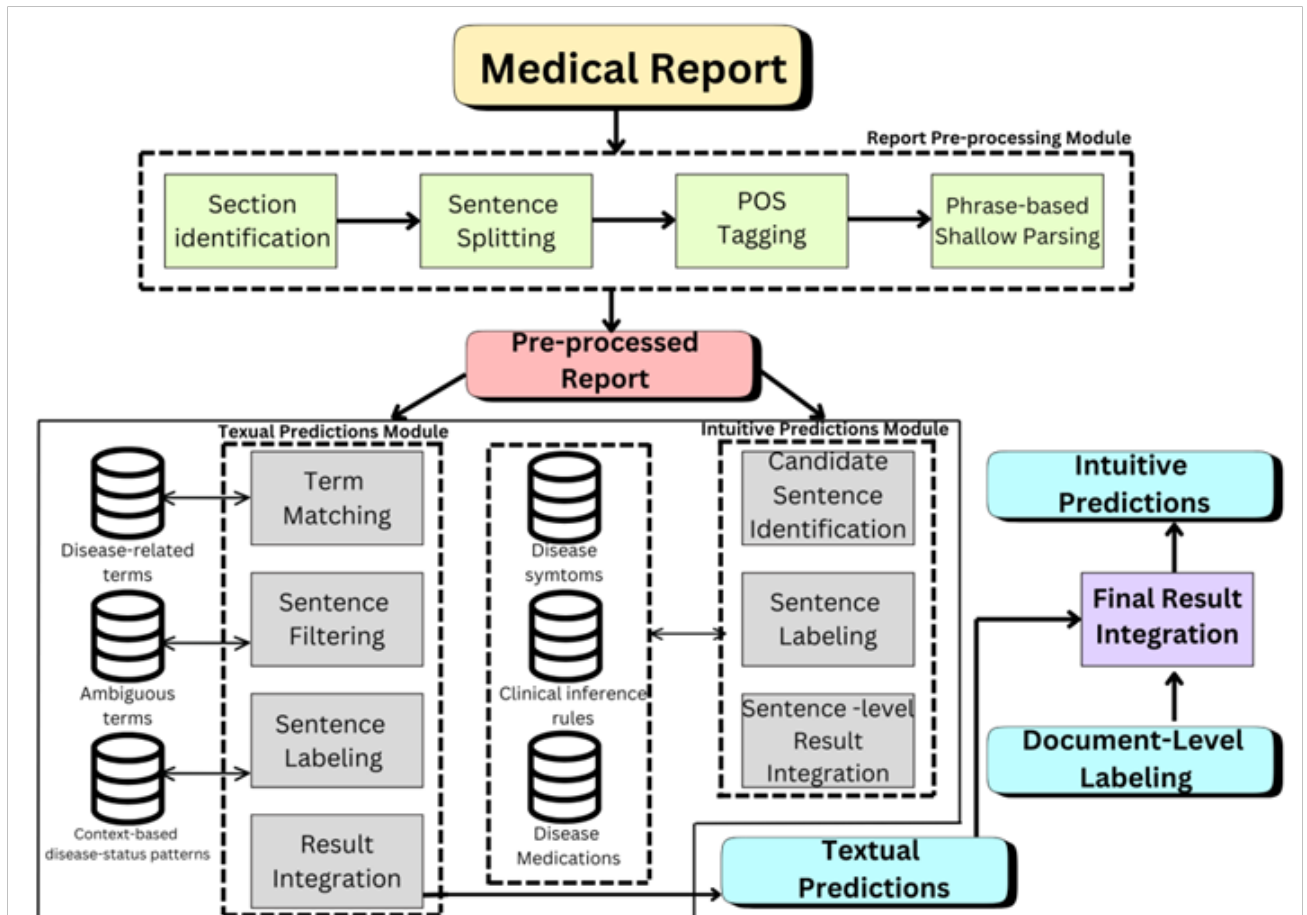


Figure 4.1: System Architecture

4.2 Data Flow Diagrams

4.2.1 Level 0 DFD

Level 0 : Context Diagram for Clinical Text Processing System

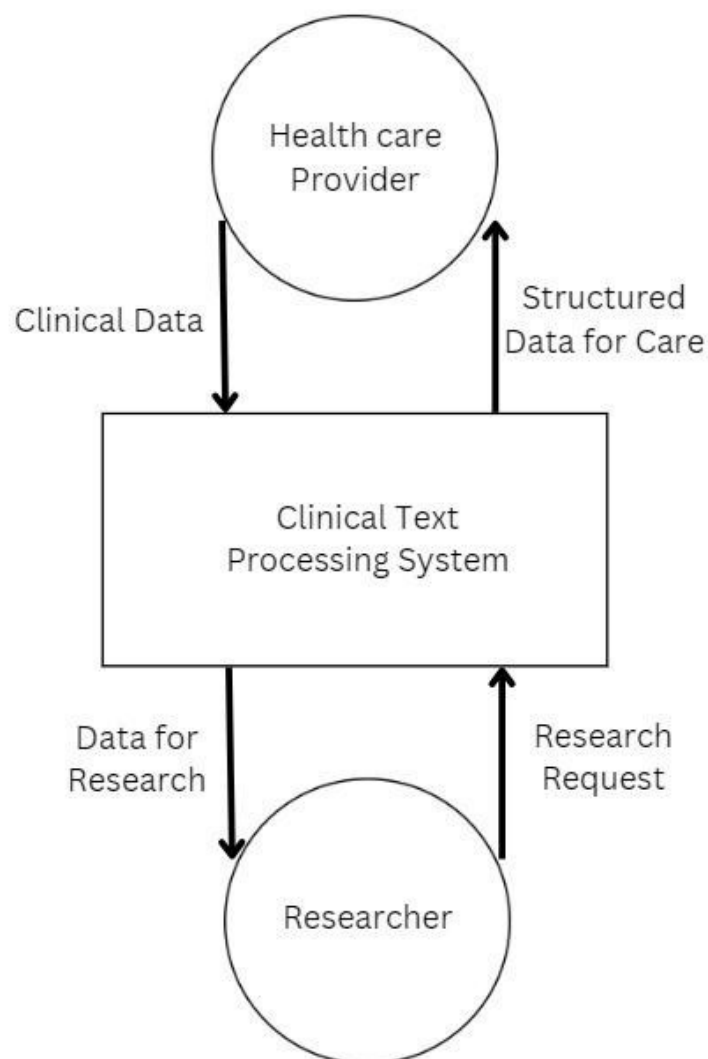


Figure 4.2: Level 0 Data Flow Diagram

4.2.2 Level 1 DFD

Level 1: Data Flow Diagram for Clinical Text Processing System

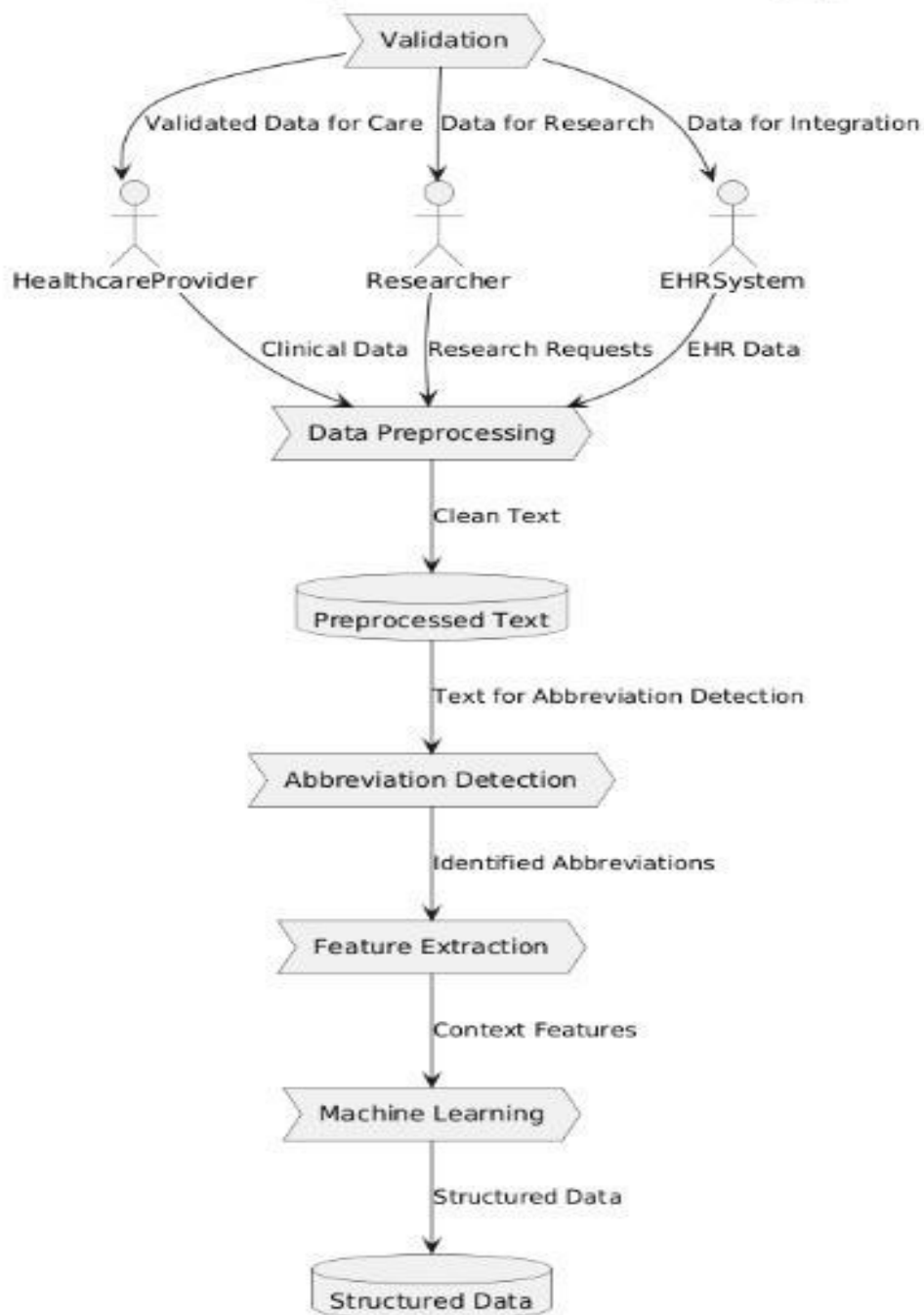


Figure 4.3: Level 1 Data Flow Diagram

4.2.3 Level 2 DFD

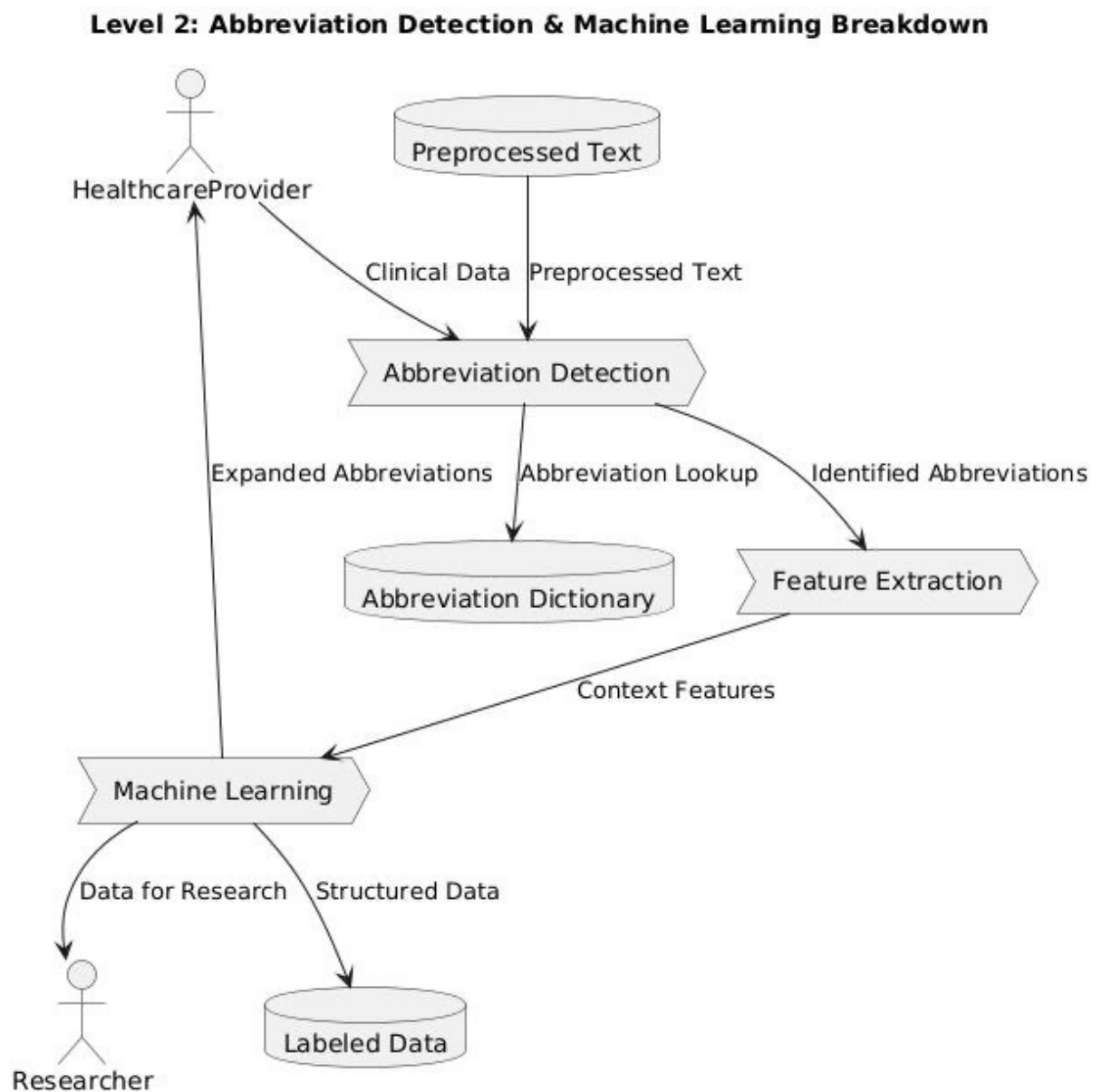


Figure 4.4: Level 2 Data Flow Diagram

4.3 Entity Relationship Diagram

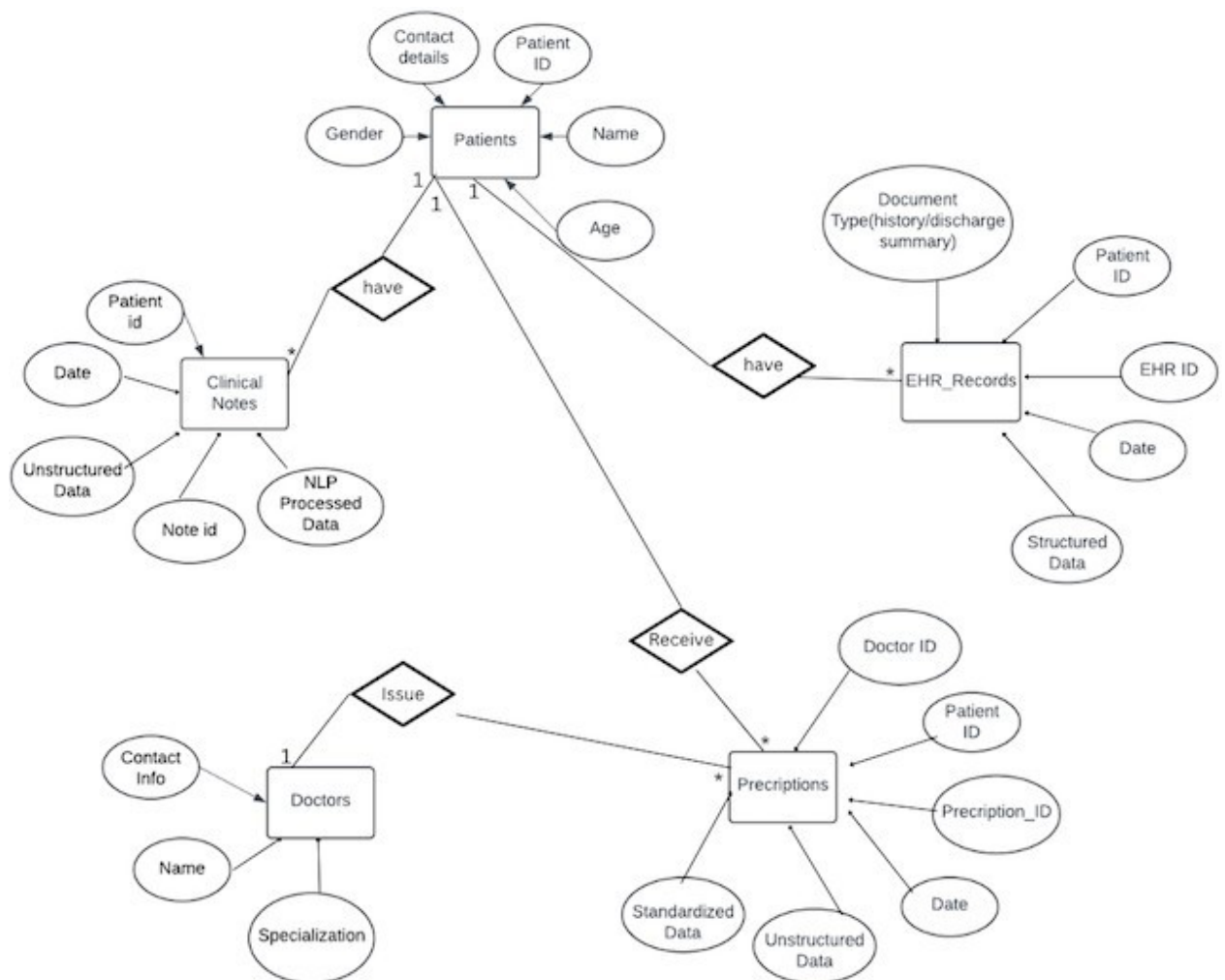


Figure 4.5: Entity Relationship diagram

4.4 UML Diagrams

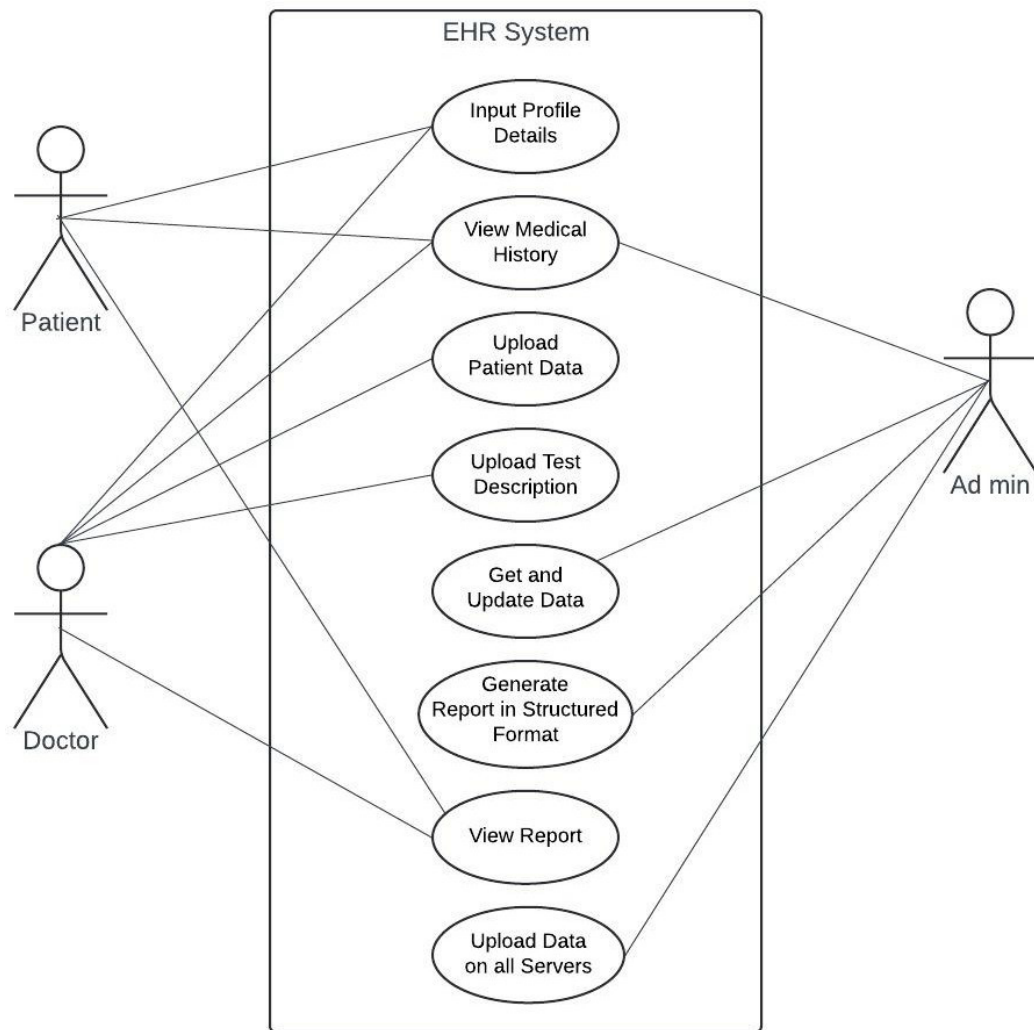


Figure 4.6: Use case diagram

Unlocking Clinical Data: Structuring and Summarizing Unstructured Medical Records

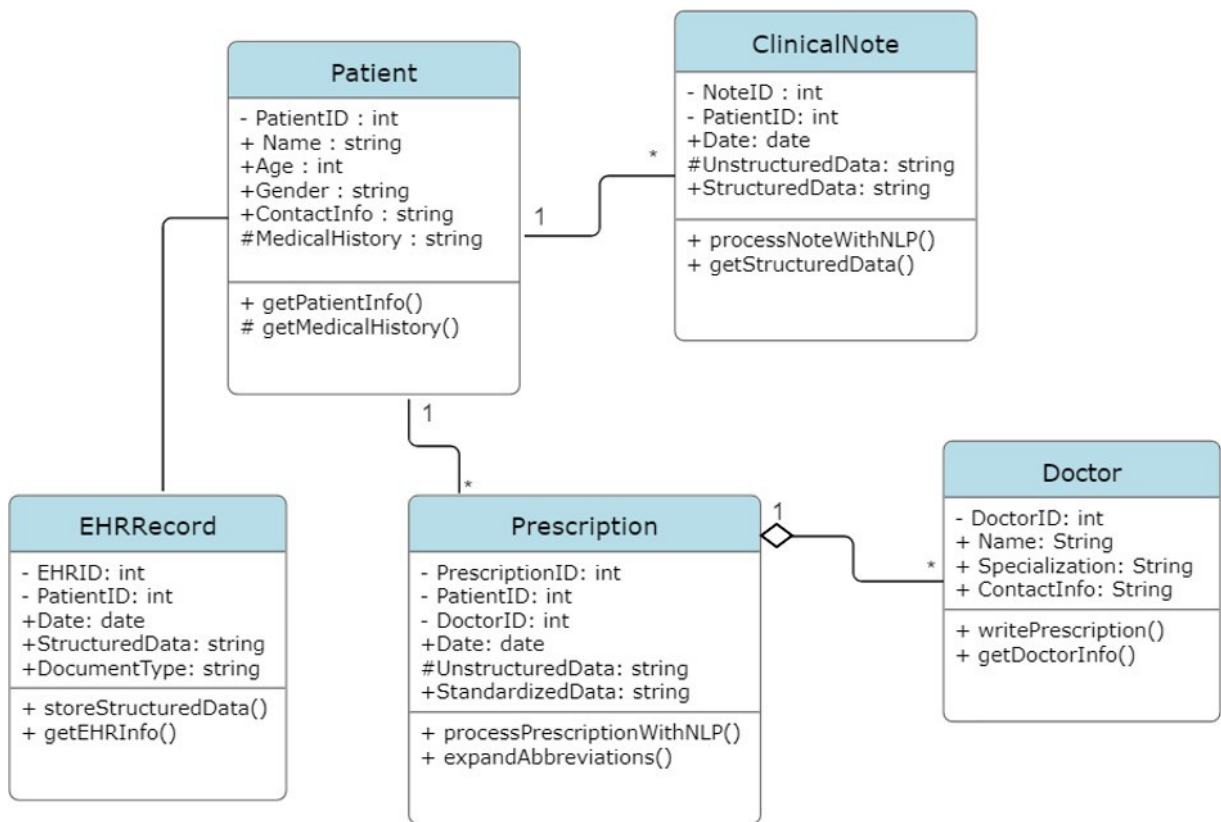


Figure 4.7: Class Diagram

4.5 Sequence Diagram

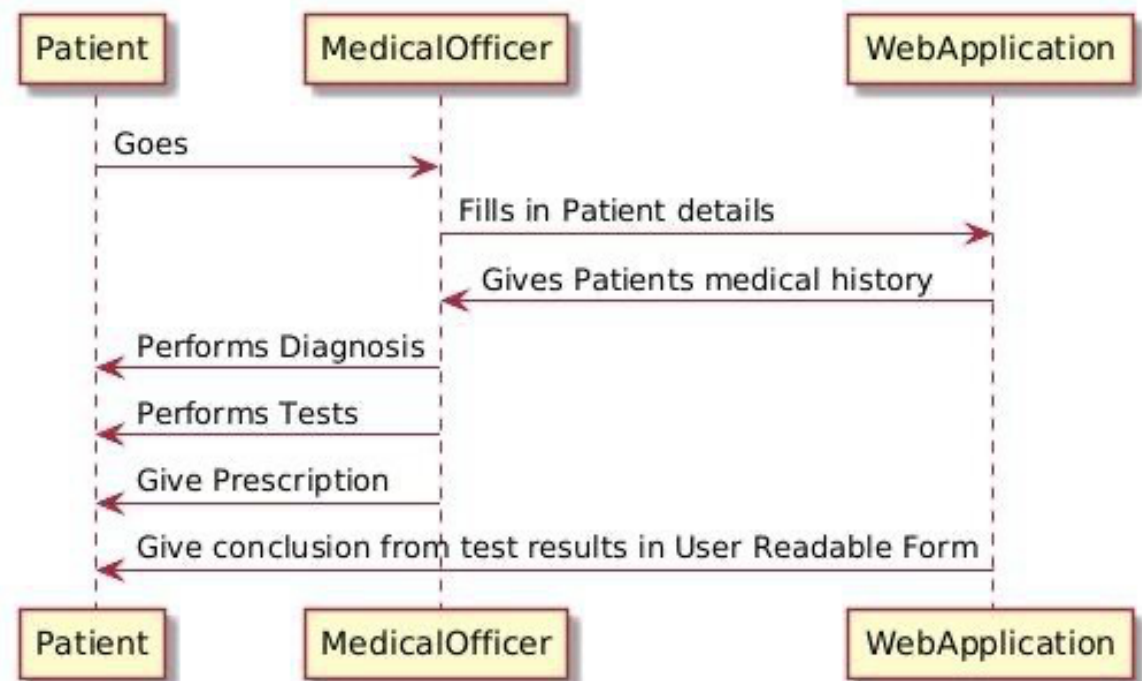


Figure 4.7: Sequence Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

The schedule and resource allocation have been somewhat adjusted as a result of our deeper examination of the clinical datasets, project complexity, and other risk factors. The project timetable has been somewhat extended to 24 weeks due to the increased complexity brought on by differences in the usage of abbreviations and possible delays in obtaining high-quality EHR data.

Data collection and preprocessing (5 weeks) :

The longer timeline accounts for the range of data types and the additional time required for data standardisation and cleansing.

Feature Extraction and Model Development (9 Weeks):

Because it is challenging to discern between medical abbreviations and the computational requirements of training deep learning models like BERT, the development phase has been extended by one week.

Model Evaluation and Refinement (8 weeks):

Feedback from healthcare professionals is expected to be more iterative, which will require additional time for model retraining and fine-tuning, extending this phase by one week.

5.1.2 Project Resources

- **Team:** 4 Computer Science students (roles rotated)
- **Hardware:** 8 GB RAM, 2.4 GHz processor, 50 GB storage
- **Software:** Google Collab (Free)
- **Libraries:** NumPy, Pandas, collections, statistics, HuggingFace Transformers, PyTorch or TensorFlow

5.2 Risk Management

5.2.1 Risk Identification

In designing an NLP-based framework for structuring unstructured biomedical data, numerous risks could affect the project's success and efficiency:

- **Data Security and Privacy Issues:** Sensitive data is used in EHRs for patient information, which could lead to privacy violations. Maintaining patient trust and avoiding legal problems need adherence to healthcare data privacy laws, such as GDPR in the EU or HIPAA in the US.
- **Inconsistent Data Quality:** There are wide variations in the terminology and structure employed in clinical data. Variations in medical records, handwritten notes, and inconsistent abbreviations can all lead to errors in data extraction and structure.
- **Restricted Access to High-Quality Healthcare Data:** The availability of sufficient high-quality data for NLP model training and testing may be restricted by institutional constraints, privacy regulations, or the proprietary nature of healthcare data.
- **The Algorithm Bias:** Developing biases through training NLP and ML models on sparse or unrepresentative datasets may result in inaccurate patient data interpretation, which could lower the standard of care.
- **Model Interpretability and Validation:** It can be challenging to make sure that the results of complex deep learning models are trustworthy, comprehensible, and validated for use in clinical settings because these models may lack transparency.

5.2.2 Risk Analysis

Every risk that has been identified is examined for chance of occurrence, potential impact on project outcomes, and viable mitigation strategies:

- **Security and Privacy of Data:** High probability, high impact. Any infraction or disobedience could lead to major ethical and legal issues. Strong data encryption, data anonymization, and access restriction are required.
- **Inconsistent Data Quality:** Medium likelihood, high impact. Inaccurate data processing might reduce usability. NLP preprocessing methods like standardization, tokenization, and abbreviation expansion can help reduce this.

- **Restricted Data Access:** Medium probability, medium impact. Limited data availability may hinder model testing and training. Using publicly available medical datasets and collaborating with healthcare organizations can help address this.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

A thorough plan consists of the following to manage and reduce the risks that have been identified:

- **Security and Privacy of Data:** Put in place stringent data anonymization and access control measures. Standards for encryption and frequent audits will be implemented to ensure data confidentiality.
- **Data Quality:** At each stage of the preprocessing process, validate and inspect the data. Standardize acronyms and language to cut down on inconsistencies.
- **Data Access:** Use open-access datasets whenever possible, and work with healthcare providers to establish data access agreements. Continuous improvement and data collection guarantee model flexibility.
- **Bias Monitoring:** Test the model on a variety of datasets on a regular basis to look for any potential biases. If inconsistencies are discovered, make the necessary corrections.
- **Model Interpretability:** Choose models with transparent or interpretable output layers whenever feasible. Monitoring the model's performance on a regular basis ensures that the results satisfy clinical requirements, increasing user confidence and practicality in a medical setting.

5.3 Project Schedule

5.3.1 Project Task Set

To guarantee a systematic progression from data collection to deployment, the project is broken up into stages, each of which has distinct tasks to complete:

1. Gathering and Preparing Data
 - Gather clinical text data, such as EHR data and clinical notes.
 - Clean text data by eliminating extraneous elements like punctuation and special characters.
 - Convert text data into words, phrases, and acronyms via tokenization.
2. Create or incorporate a thorough abbreviation dictionary to set up dictionary lookup.
 - Use morphology-based matching to identify patterns in abbreviations.
3. Contextual Extraction of Features
 - Analyze surrounding text to find abbreviations based on context.
 - Use NLP techniques like dependency parsing and POS tagging to extract syntactic and semantic elements.
4. Disambiguation of Abbreviations (Machine Learning Method)
 - Add enlarged versions of abbreviations to training data.
 - Create features such as word embeddings, sentence structure, and contextual windows.
 - Classifiers such as Naive Bayes, SVM, or deep learning models like LSTMs or BERT can be used to train the machine learning model.
5. Fallback Using a Dictionary
 - For unclear situations, use dictionary lookup as a fallback method.
6. Finalization and Verification
 - Create a validation system for important abbreviation expansions with feedback from medical experts.
7. Assessment
 - Establish criteria for assessing the system, such as precision, recall and F1-score.
 - To guarantee generalization across different kinds of clinical notes, conduct cross-validation.

8. Implementation

- Incorporate the model into the electronic health record.
- After deployment, keep an eye on system performance and make any required modifications.

5.3.2 Task Network

Workflows are set up to balance parallel and sequential processes:

- Sequential Dependencies: Before detecting acronyms, data collection and preprocessing must be finished. Contextual feature extraction is also necessary for the completion of abbreviation disambiguation.
- Parallel Tasks: Contextual feature extraction, morphology-based matching, and dictionary lookup can all take place at the same time.
- Integration Points: Dictionary-Based Fallback and Post-processing will combine into a single system for validation and deployment following the conclusion of model training.

5.3.3 Timeline Chart

Task	Timeline	Duration
Initial Research	12 July – 31 Aug	14 days
Dataset Collection	1 Aug – 15 Sept	32 days
Analyzing the previous Research	16 Sept – 10 Oct	19 days
Write and finalize the review paper for submission.	11 Oct – 25 Oct	11 days
Define Project Scope, Finalize Scope, and select preferred algorithms for NLP and ML implementation.	26 Oct – 10 Nov	10 days
Preprocessing the Dataset	11 Nov – 30 Dec	36 days
Model Training, Testing and Tuning	1 Jan – 15 Feb	33 days
Evaluation of Result and refinement	16 Feb – 25 Feb	7 days
Finalize project report and prepare for presentation.	26 Feb – 5 Mar	6 days

Tab 5.1: Timeline Chart

5.4 Team Organization

5.4.1 Team structure

- **Member 1 – Data and NLP Lead:** Handled dataset cleaning, preprocessing, and implemented the abbreviation disambiguation module using BERT and a custom dictionary.
- **Member 2 – Summarization and Model Trainer:** Trained and fine-tuned the Pegasus model for clinical summarization, managed tokenization, batching, and checkpointing.
- **Member 3 – UI Developer and Integrator:** Developed the user interface, integrated all system components, and managed project coordination and documentation.
- **Member 4 – Evaluation and Documentation Lead:** Evaluated model performance, analyzed output quality, and contributed to writing the paper's conclusion, future work, and applications.

5.4.2 Management reporting and communication

- **Weekly Meetings:** Internal team meetings were conducted every week about an 1 hour to review progress, discuss challenges, and plan next steps.
- **Progress Reporting:** Bi-weekly progress updates were shared with the department to ensure transparency and alignment with academic requirements.
- **Communication Tools:** A WhatsApp group was maintained for daily coordination and quick queries, while Google Meet was used for all virtual discussions.
- **Development and Documentation Tools:** Google Collab was used for implementing and testing models, while Google Docs was used for collaborative documentation and report writing.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

1. Data Generation & Preprocessing Module

Purpose: Generate synthetic EHR records using Faker with realistic fields like vitals, symptoms, diagnosis.

Tasks:

- Generate diverse patient profiles
- Format textual and numerical attributes
- Create rich descriptions for summarization

Output: Structured dataset (CSV/DF) ready for ML pipeline the data is structured for effective experimentation and analysis.

2. Text Construction Module

Purpose: Compose detailed, medically-relevant text from tabular data.

Tasks:

- Combine symptoms, vitals, and diagnosis into paragraphs
- Generate Combined_Text and Detailed_Description columns.

Output: Text input suitable for feeding to NLP models.

3. Summarization Module

Purpose: Generate concise medical summaries using pretrained transformers.

Tasks:

- Load and run models like BioBERT, T5, or BioBART
- Apply summarization pipeline over descriptive columns.

Output: Human-readable summaries in the Summary column.

4. Backend API Module

Purpose: Bridge frontend and ML logic with clean, RESTful API endpoints.

Tasks:

- Accept input data via POST requests
- Process and return summaries
- Expose model predictions

Framework: Fast API

5. Frontend UI Module

Purpose: Provide an interface to upload EHR entries and view summaries.

Tasks:

- Form for patient data submission

- Display original and summarized reports

Tech: React.js

6. Abbreviation & Disambiguation Module

Purpose: Detect and expand abbreviations, resolve ambiguous medical terms.

Tasks:

- Map terms like "BP" to "Blood Pressure"
- Use context to resolve terms like "MI" (Myocardial Infarction vs Mitral Insufficiency)

7. Database & Storage Module

Purpose: Store EHRs, summaries, user interactions.

Tasks:

- Save patient entries and their generated summaries
- Manage session history and access controls

6.2 Tools and Technologies Used

Programming Language

- **Python** – Used across all modules for data handling, processing, and ML implementation

Machine Learning & NLP Frameworks

- **Transformers (Hugging Face)** – For advanced NLP tasks including summarization and abbreviation resolution
- **spaCy** – Lightweight NLP library for tokenization, POS tagging, and entity recognition
- **Scikit-learn** – For building custom classifiers (e.g., context-based abbreviation disambiguation)

Data Handling

- **Pandas** – For tabular data manipulation and abbreviation context extraction
- **Faker** – Used to generate mock patient data including domain-relevant abbreviations

Abbreviation Disambiguation

- **BioBERT** – Fine-tuned for medical language understanding and abbreviation expansion
- **MedMentions Dataset / UMLS (Unified Medical Language System)** – Used as reference data for identifying and mapping abbreviations to correct terms
- **Custom Contextual Embeddings** – To resolve abbreviation meanings based on sentence-level semantics

Summarization Models

- **T5 / BioBERT / BioBART** – For summarizing clinical notes with resolved abbreviations

Web Integration

- **Flask / FastAPI** – For serving API endpoints to the frontend for summarization and abbreviation resolution

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- **React.js** – User interface for inputting records and viewing disambiguated summaries

Storage & Retrieval

- **CSV / SQLite / PostgreSQL** – Storage of raw and processed data, resolved forms, and summary results

Utilities & Deployment

- **Google Colab / Jupyter Notebook** – Prototyping and testing disambiguation logic
- **GitHub**-Version control

6.3 Algorithm Details

6.3.1 Algorithm 1: Medical Text Summarization using Transformer-based Models

Objective:

To summarize complex medical narratives into compact, clinically coherent text for rapid understanding.

Steps:

1. **Text Preprocessing:**
 - Clean input (remove extra spaces, convert to lowercase)
 - Normalize special medical characters and formatting
2. **Tokenizer Encoding:**
 - Use PEGASUS tokenizer to convert input into model-friendly format
 - Truncate or pad sequences to match model input size
3. **Inference:**
 - Pass tokens into PEGASUS model (fine-tuned on medical summarization, if available)
 - Generate summary using beam search or sampling with temperature controls
4. **Postprocessing:**
 - Capitalize sentences, correct grammar
 - Verify critical medical terms are retained (using a medical NER filter, optional)
5. **Result Handling:**
 - Store or render the summarized content on the frontend

6.3.2 Algorithm 2: Abbreviation Disambiguation using BERT

Objective:

To detect and correctly expand ambiguous medical abbreviations based on context.

Steps:

1. **Abbreviation Extraction:**
 - Use regular expressions and medical lexicons (e.g., ADAM, UMLS) to detect candidate abbreviations.
2. **Context Window Retrieval:**
 - Capture ± 5 tokens around the abbreviation in question
3. **Contextual Embedding with BERT:**
 - Use BioBERT or ClinicalBERT to convert context into a dense vector representation
4. **Candidate Expansion Generation:**
 - Retrieve possible full forms for the abbreviation from a medical abbreviation dictionary.
5. **Similarity Scoring:**
 - Generate embeddings for each full form using the same BERT model
 - Compute cosine similarity between context and each candidate
6. **Disambiguation Decision:**
 - Select the full form with the highest score
 - If score falls below a threshold, mark as uncertain or request user validation
7. **Replacement & Logging:**
 - Replace abbreviation in original text with the selected full form
 - Optionally log the abbreviation and its resolved form for audit

CHAPTER 7

SOFTWARE TESTING

Unlocking Clinical Data: Structuring and Summarizing Unstructured Medical Records

7.1 Test cases and results

Training fold 1

Some weights of BertForTokenClassification were not initialized from the model checkpoint at emilyalsentzer/Bio_ClinicalBERT and are newly initialized: ['classifier.bias']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

[355/355 1:33:38, Epoch 4/5]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	2.216500	2.131178	0.887784	0.837198	0.887784	0.855730
2	0.575600	0.821271	0.950523	0.943917	0.950523	0.944379
3	0.388800	0.480420	0.969396	0.961040	0.969396	0.963081
4	0.295200	0.430952	0.976792	0.974196	0.976792	0.973304

[36/36 01:02]

Fold 1 evaluation results: {'eval_loss': 0.43095216155052185, 'eval_accuracy': 0.9767916347870441, 'eval_precision': 0.9741957497118338, 'eval_recall': 0.9767916347870441}

Training fold 2

Some weights of BertForTokenClassification were not initialized from the model checkpoint at emilyalsentzer/Bio_ClinicalBERT and are newly initialized: ['classifier.bias']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

[355/355 1:38:20, Epoch 4/5]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	2.257000	2.045072	0.902210	0.849910	0.902210	0.873719
2	0.686700	0.685413	0.953868	0.952614	0.953868	0.950038
3	0.300000	0.460807	0.970447	0.969413	0.970447	0.967770
4	0.293600	0.443028	0.975733	0.972688	0.975733	0.972056

[36/36 01:17]

Training fold 3

Some weights of BertForTokenClassification were not initialized from the model checkpoint at emilyalsentzer/Bio_ClinicalBERT and are newly initialized: ['classifier.bias']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

[355/355 2:13:47, Epoch 4/5]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	2.263900	2.031049	0.881744	0.834086	0.881744	0.854592
2	0.680400	0.698086	0.958118	0.951363	0.958118	0.952000
3	0.331800	0.441850	0.975610	0.970728	0.975610	0.970934
4	0.235500	0.442994	0.977581	0.978462	0.977581	0.975236

[36/36 01:22]

Fold 3 evaluation results: {'eval_loss': 0.44299396872520447, 'eval_accuracy': 0.9775806848977581, 'eval_precision': 0.9784616787861912, 'eval_recall': 0.9775806848977581}

Training fold 4

Some weights of BertForTokenClassification were not initialized from the model checkpoint at emilyalsentzer/Bio_ClinicalBERT and are newly initialized: ['classifier.bias']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

[355/355 2:55:59, Epoch 4/5]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	2.258000	2.143884	0.887528	0.829219	0.887528	0.855178
2	0.723000	0.802253	0.945264	0.942957	0.945264	0.939734
3	0.311300	0.497420	0.962259	0.957440	0.962259	0.957210
4	0.271600	0.405001	0.974256	0.971903	0.974256	0.970103

[36/36 01:08]

Fold 4 evaluation results: {'eval_loss': 0.405000776052475, 'eval_accuracy': 0.9742564358910273, 'eval_precision': 0.9719027868498757, 'eval_recall': 0.9742564358910273,}

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Fold 4 evaluation results: {'eval_loss': 0.405000776052435, 'eval_accuracy': 0.9742564358918273, 'eval_precision': 0.9719027868408757, 'eval_recall': 0.9742564358918273, 'eval_f1': 0.9730111111111111},
Training fold 5
Some weights of BertForTokenClassification were not initialized from the model checkpoint at emilyalsentzer/Bio_ClinicalBERT and are newly initialized: ['classifier.bias']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

[335/335 1:21:55. Epoch 4/5]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	2.348600	2.125247	0.884545	0.832206	0.884545	0.853011
2	0.766500	0.757679	0.960903	0.954807	0.960903	0.956796
3	0.307600	0.479290	0.971399	0.969198	0.971399	0.967682
4	0.237200	0.383467	0.975597	0.979018	0.975597	0.973736

[36/36 01:13]

Fold 5 evaluation results: {'eval_loss': 0.3934665024280548, 'eval_accuracy': 0.9755969561794805, 'eval_precision': 0.9790177824162372, 'eval_recall': 0.9755969561794805, 'eval_f1': 0.9770111111111111},

[36/36 01:13]

CHAPTER 8

RESULTS

8.1 Outcomes

Test metrics:

```
{'eval_accuracy': 0.9954792043399638,  
 'eval_f1': 0.9957553010893057,  
 'eval_loss': 0.07595796138048172,  
 'eval_model_preparation_time': 0.0067,  
 'eval_precision': 0.9969249335068233,  
 'eval_recall': 0.9954792043399638,  
 'eval_runtime': 15.145,  
 'eval_samples_per_second': 3.301,  
 'eval_steps_per_second': 0.462}
```

```
# Test predictions  
# Sample note from test dataset  
sample_note = test_df.iloc[15]["Original Note"]  
predictions = predict_abbreviations(sample_note)  
print(f"Sample note: {sample_note}")  
print(f"Predictions: {predictions}")
```

```
Raw predictions: [39, 39, 39, 39, 39, 39, 39, 39, 39, 26, 26, 39, 39, 39, 39, 39, 39, 39, 39, 39, 39, 21, 21, 39, 39, 39, 12, 39, 39, 39, 39, 39, 39]  
Token: A, Label ID: 39, Label: 0  
Token: 70-year-old, Label ID: 39, Label: 0  
Token: male, Label ID: 39, Label: 0  
Token: with, Label ID: 39, Label: 0  
Token: PC, Label ID: 26, Label: PCProstateCancer  
Token: presented, Label ID: 39, Label: 0  
Token: for, Label ID: 39, Label: 0  
Token: follow-up., Label ID: 39, Label: 0  
Token: Imaging, Label ID: 39, Label: 0  
Token: showed, Label ID: 39, Label: 0  
Token: no, Label ID: 39, Label: 0  
Token: progression., Label ID: 39, Label: 0  
Token: BP, Label ID: 21, Label: BPPloodPressure  
Token: was, Label ID: 39, Label: 0  
Token: normal., Label ID: 39, Label: 0  
Token: PT, Label ID: 12, Label: PTPhysicalTherapy  
Token: was, Label ID: 39, Label: 0  
Token: advised, Label ID: 39, Label: 0  
Token: for, Label ID: 39, Label: 0  
Token: mobility., Label ID: 39, Label: 0  
Sample note: A 70-year-old male with PC presented for follow-up. Imaging showed no progression. BP was normal. PT was advised for mobility.  
Predictions: [('PC', 'PCProstateCancer'), ('BP', 'BPPloodPressure'), ('PT', 'PTPhysicalTherapy')]
```

8.2 System Screen Shots

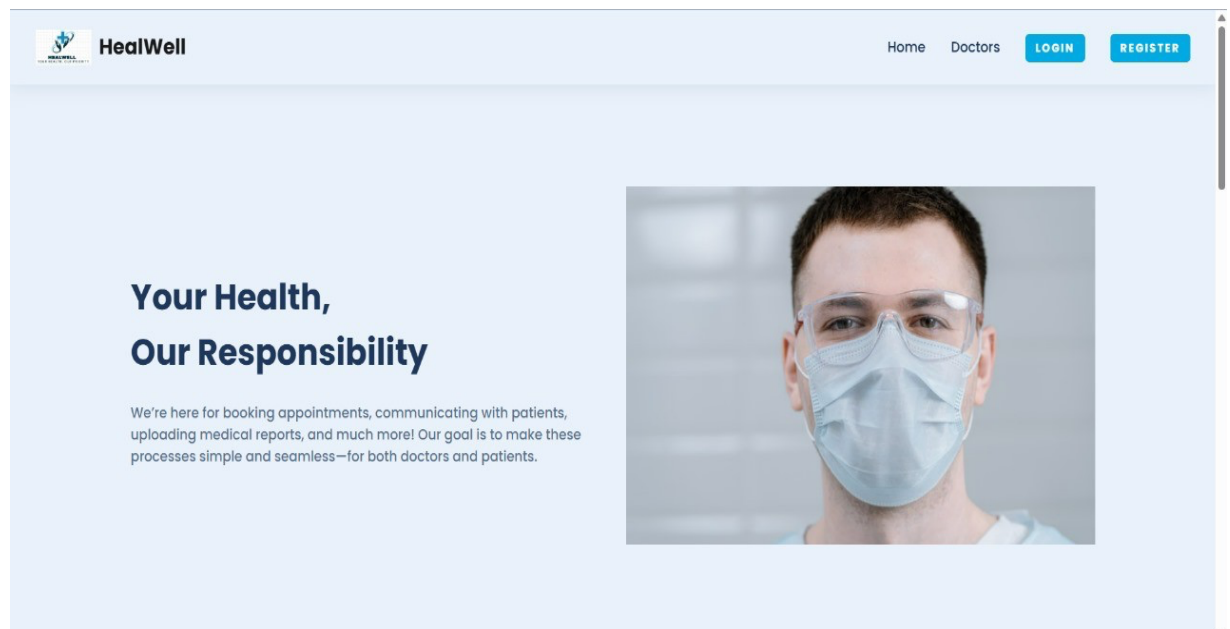


Fig 8.1: System UI

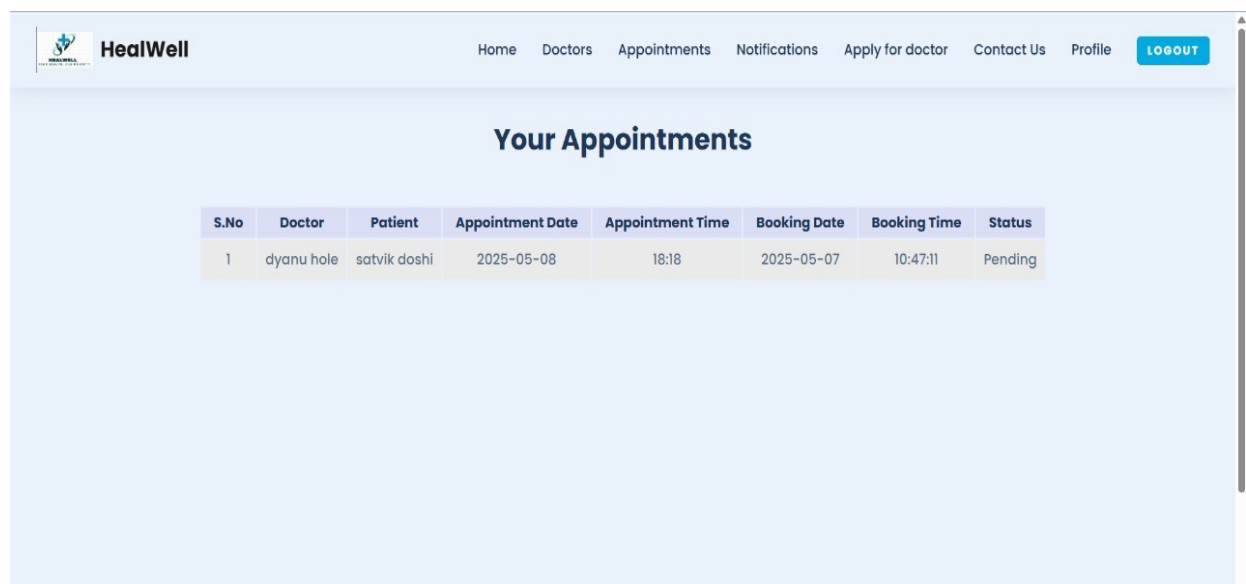


Fig 8.2: System UI

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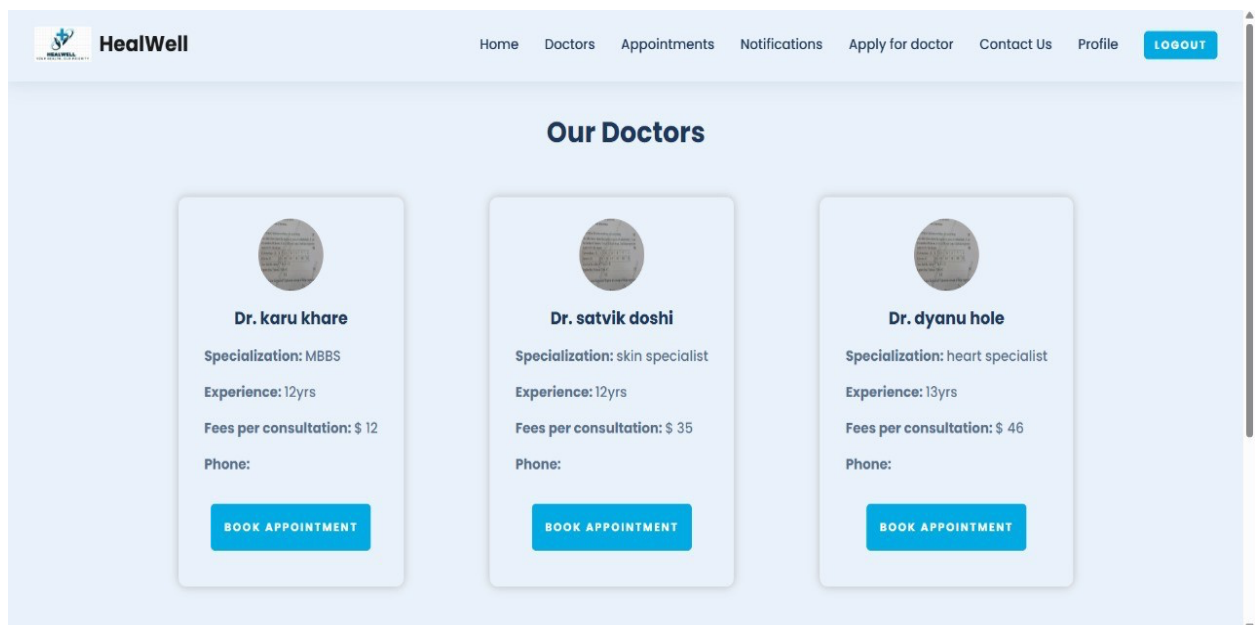


Fig 8.3: System UI

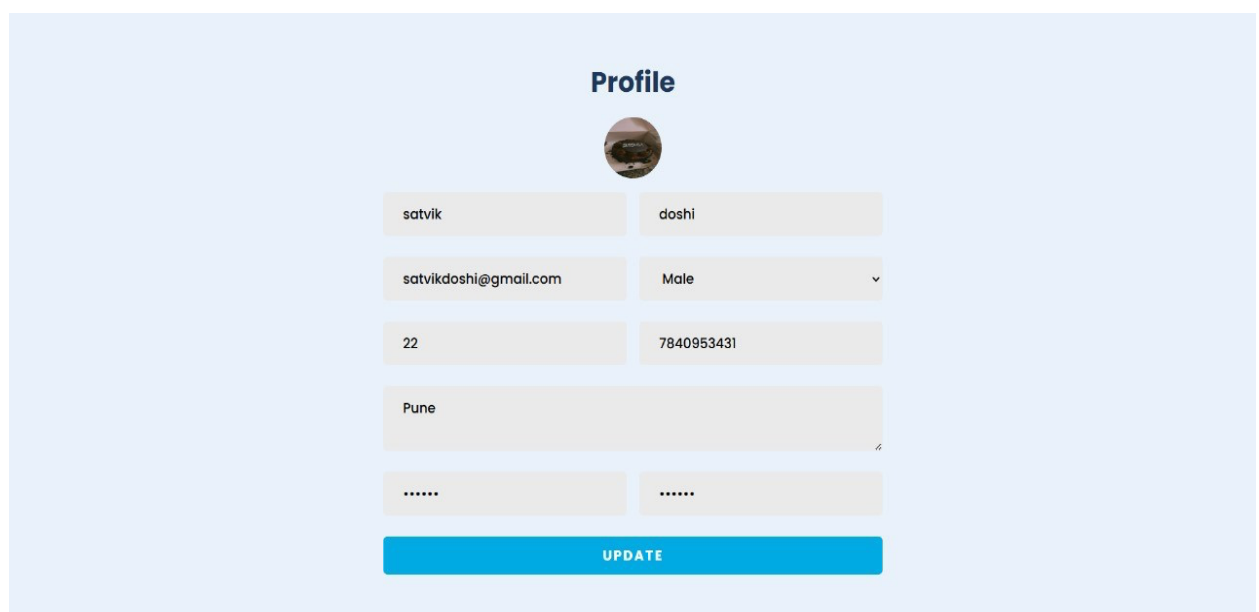


Fig 8.4: System UI

CHAPTER 9

CONCLUSIONS

9.1 Conclusion

In this study, we developed a system that can both disambiguate abbreviations and summarize medical texts in order to address the problem of processing unstructured clinical data within Electronic Health Records (EHRs). Leveraging the deep contextual understanding of transformer-based model BERT, we accurately identified, interpreted, and expanded medical abbreviations such as "RA" and "HR" based on surrounding context. Additionally, the technology generated clear, concise overviews of lengthy clinical documents, assisting medical practitioners in making quicker and better decisions.

Our method outperformed baseline models in terms of summarization quality, contextual relevance, and medical accuracy. By providing a scalable and dependable method for improving the interpretability of unstructured EHR data, the presented system makes a substantial contribution to the field of biomedical natural language processing.

9.2 Future Work

Even if the current implementation performs well, there are a few improvements that could increase the system's impact and adaptability:

1. **Clinical Decision Support:** By providing concise descriptions and straightforward term expansion, this tool helps physicians rapidly comprehend complicated medical histories.
2. **Expanded Domain Coverage:** Inclusion of domain-specific abbreviations from specializations like oncology, pediatrics, and neurology to broaden applicability.
3. **Multilingual Support:** Expanding the ability to decipher and summarize abbreviations in non-English and regional languages to accommodate a range of healthcare environments.
4. **User Feedback Loop:** Allowing for ongoing model improvement by integrating clinician input to increase summarization accuracy and abbreviation mapping.
5. **Advanced Domain-Specific Models:** To improve semantic understanding even more, domain-tuned BioBERT variants, ClinicalBERT, or Med-BERT may be included.

Implementation in Clinical Settings: Working with medical facilities to verify and improve the system with actual patient data in clinical settings.

9.3 Applications

The developed system has applications in various fields of biomedical research and healthcare:

1. **Electronic Health Record (EHR) Optimization:** Enhances readability and structure of clinical notes, aiding both healthcare providers and administrative staff.
2. **Clinical Decision Support:** By providing concise descriptions and straightforward term expansion, this tool helps physicians rapidly comprehend complicated medical histories.
3. **Medical research:** Makes it easier to extract structured data for analytics, trend analysis, and disease prediction in extensive investigations.
4. **Medical Transcription Systems:** Improves transcription accuracy by resolving ambiguous abbreviations before storage or analysis.
5. **Training and Education:** By making shorthand easier to understand and offering palatable summaries of actual situations, it serves as a useful tool for training medical students or new hires.
6. **Healthcare Interoperability:** Promotes standardization and clarity in cross-system data exchange, essential for collaborative or remote care.

CHAPTER 10
APPENDIX

10.1 Appendix A

Problem Statement Feasibility Assessment

1. Problem Statement Analysis

- **Satisfiability Analysis:**

- Abbreviation detection is complicated in the circumstances of ambiguity but is satisfied by dictionary lookups and morphology-based matching.
- Contextual Expansion: Difficult with minimal context in clinical notes, but satisfactory with correct feature extraction.

2. Computational Complexity

We analyze the problem's complexity classification under NP-Hard, NP-Complete, or P-Type based on its components:

- **P-Type:** Basic feature extraction and abbreviation detection can be solved in poly-nomial time.
- **NP-Complete:** Because of ambiguity, context-based abbreviation disambiguation is similar to NP-Complete problems.
- **NP-Hard:** Because of its great complexity, using deep learning models for real time disambiguation makes the task NP-Hard.

3. Modern Theory Application

- **Group Theory:** Used to arrange expansions and acronyms.
- Clinical notes can be modelled as graphs using graph theory to aid in context-based disambiguation.

4. Conclusion: While disambiguation might result in NP-complete or NP-Hard circumstances, particularly with deep learning models, abbreviation detection is possible in polynomial time. Although algorithmic frameworks give things structure, they can make computation more difficult.

5. Implementation: To address the challenges outlined above, we implemented a solution using a BERT-based model for abbreviation detection

and disambiguation in clinical notes. The implementation details are as follows:

- **Model Architecture:** We used Bio_ClinicalBERT, a pre-trained BERT model fine-tuned for clinical text, as the base model. A token classification head was added to predict labels for each token, where labels correspond to specific abbreviations (e.g., CHFCongestiveHeartFailure, AFAFAtrialFibrillation) or 'O' for non-abbreviations. The model was fine-tuned with a custom number of labels based on a predefined abbreviation dictionary.
- **Data Preparation:** The dataset consisted of clinical notes with original and expanded forms of abbreviations. We created a labelled dataset by extracting abbreviation-full form pairs and aligning them with tokens. To address class imbalance (where 'O' labels dominated), we generated synthetic data by creating notes for each abbreviation (e.g., "Patient with CHF was treated.") and repeated this process 10 times to increase abbreviation examples. Additionally, we applied data augmentation by replacing words with synonyms (e.g., "presented" with "admitted") to improve model robustness.
- **Training Process:** The model was trained using 5-fold cross-validation to ensure robust evaluation. We implemented a custom trainer with a weighted loss function, downweighing the 'O' label (weight = 0.1) to prioritize learning abbreviation labels. Training hyperparameters included a learning rate of 5e-5, 5 epochs, 500 warmup steps, and gradient accumulation to simulate larger batch sizes. The best model was selected based on the F1 score on the validation set.
- **Evaluation and Results:** The model was evaluated on a synthetic test dataset of clinical notes. Metrics included accuracy, precision, recall, and F1 score, computed on non-ignored labels (excluding padding tokens). The model was tested on sample notes, such as "Patient presented with CHF and AF," to verify its ability to detect and disambiguate abbreviations. Debugging was performed to ensure proper labelling and data preparation, confirming that abbreviation labels were correctly assigned during training.

10.2 Appendix B

- **Survey Paper Publication Details:**

- **Paper Title:** Unlocking Clinical Data: Structuring and Summarizing Medical Records
- **Conference:** International Conference on Innovative Computing and Communication (2025)
- **Journal:** Springer



(a) Renuka Certificate



(b) Dnyaneshwari Certificate



(c) Isha Certificate

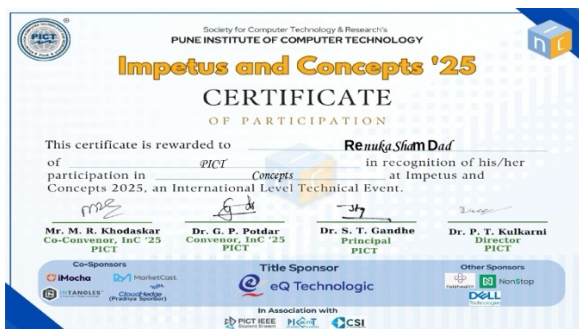


(d) Kartiki Certificate

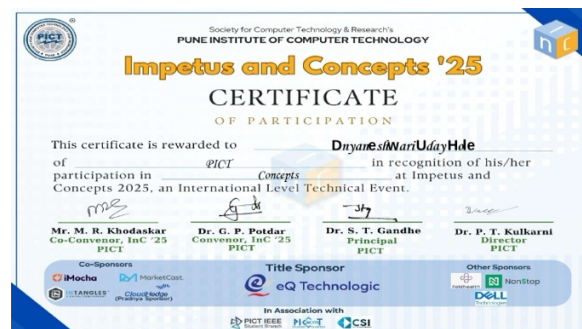
Fig 10.1: Conference Certificates

- **Event Participation Details:**

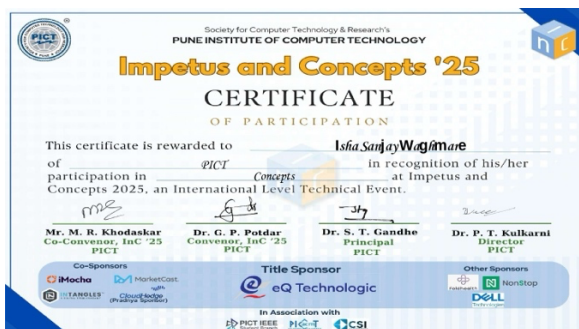
- **Event:** Impetus and Concepts- National Level Technical Symposium
- **Organizer:** Pune Institute of Computer Technology
- **Certificate of Participation:**



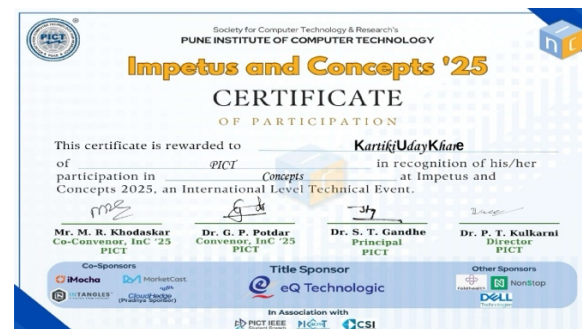
(a) Renuka Certificate



(b) Dnyaneshwari Certificate



(c) Isha Certificate



(d) Kartiki Certificate

Fig 10.2: Event Participation Certificates

10.3 Appendix C

Plagiarism Report of project report.

Document Information

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Fig 10.3: Plagiarism Report

CHAPTER 11

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
EHR	Electronic Health Record
NLP	Natural Language Processing
SVM	Support Vector Machine
BERT	Bidirectional Encoder Representations from Transformers
CRF	Case Report Form
BART	Bidirectional and Autoregressive Transformer
CORS	Cross Origin Resource Sharing
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act

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